Design of a Novel Propulsion System for a Mission to Phobos - PEPS

Spacecraft Propulsion and Design Challenge

Chair of Space Mobility and Propulsion

Group 3:

Ali Akbey

Jessica Clatza

Leonardo Eitner

Matteo Solesio

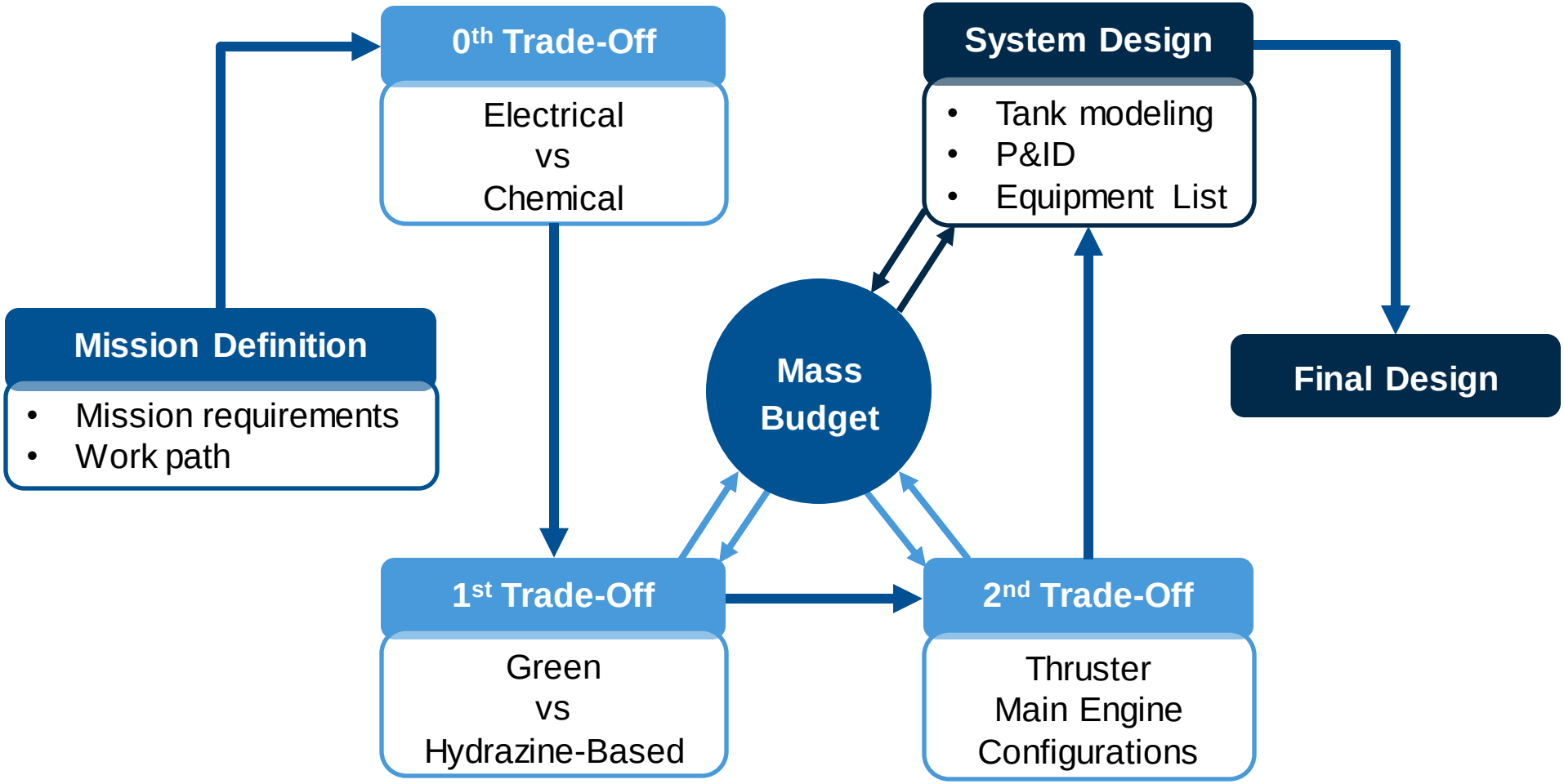


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Design Process



0th Trade-Off: Electrical vs Chemical Propulsion

Comparison criteria:

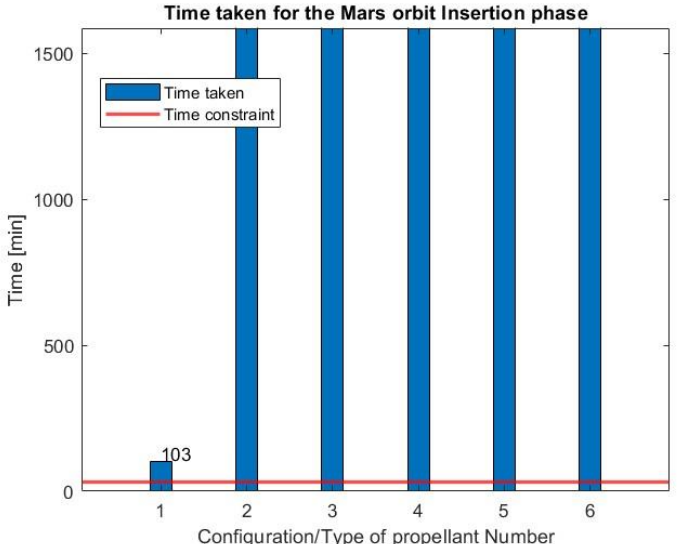
- Required time for MOI
- Functionality
- Weight

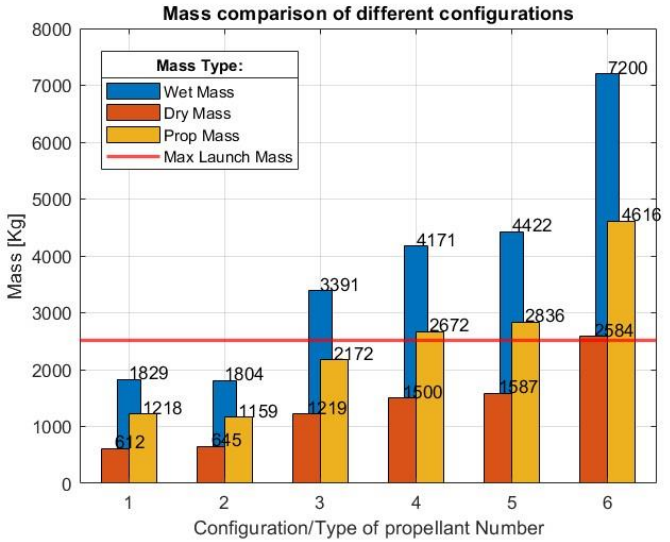
Scenarios considered:

- Main engine
- ADCS thrusters
- Both

Outcome:

No adequate thruster available COTS





1st Trade-Off: Green vs Hydrazine-Based

Trade-off goal:

Hydrazine-based Vs. Green propellants

Propellants considered:

- MMH / NTO
- Ethanol / 95% H_2O_2
- Propene / Nitrous Oxide

Criteria and weights:

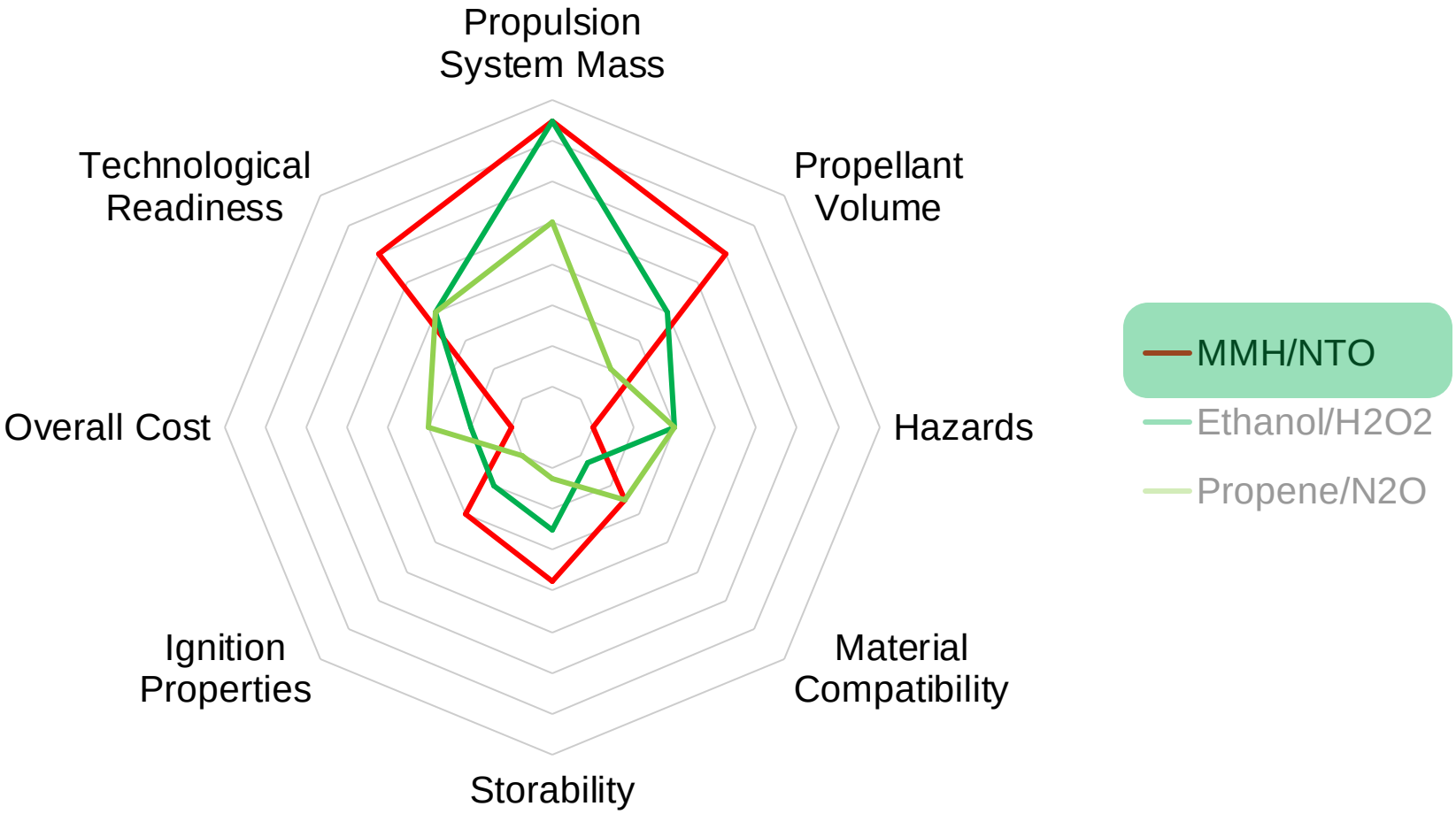
V. Bombelli et al., O. Frota et al.

Reference Mission:

MMX

Criteria	Weight
Propulsion system total mass	25%
Propellant volume	20%
Hazards	10%
Material compatibility	12,5%
Storability	12,5%
Ignition (hypergolic properties)	10%
Cost (testing, propellants, production)	10%
Technological readiness	20%

First Trade-Off: Results



2nd Trade-Off: Criteria and Weights

Trade-off goal:

Select the best main engine/thruster combination

Propellants considered:

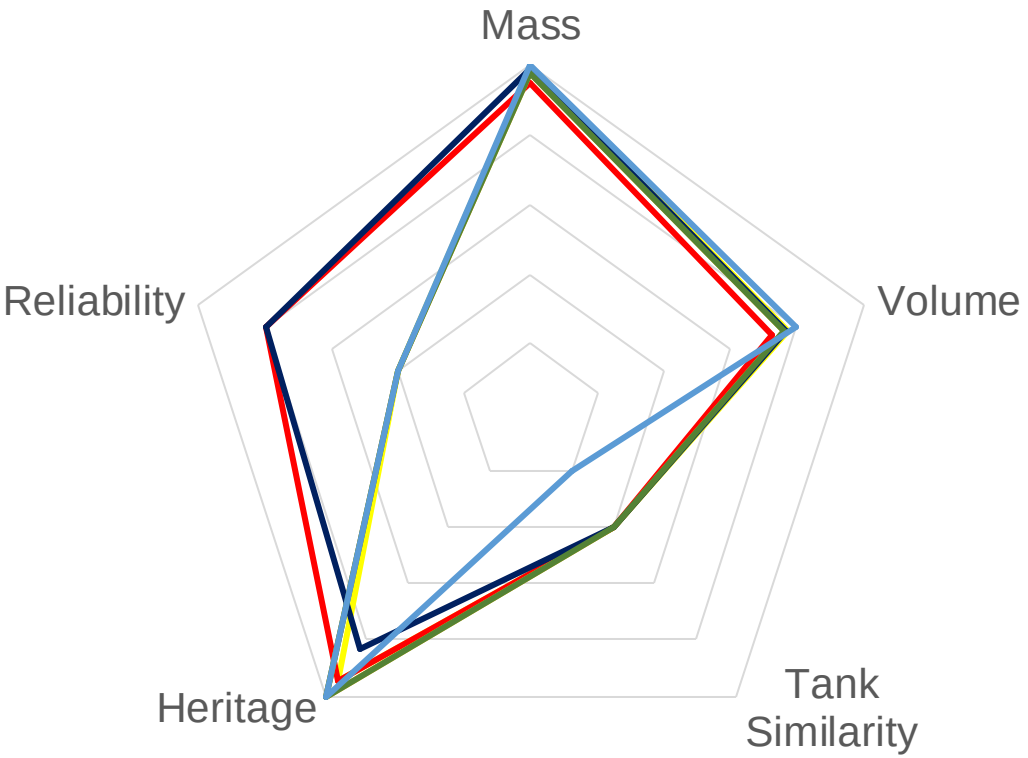
- MMH / NTO
- MMH / MON-3
- Hydrazine / MON-3

Thrust range considered:

- Main engine: 425 - 1000N
- Thrusters: 22N

Criteria	Weight
Propulsion system mass	25%
Propellant volume	20%
Tank similarity	15%
Heritage	15%
Reliability	20%

Second Trade-Off: Results



1: Medium Thrust, MMH/NTO

2: High Thrust, MMH/NTO

3: Medium Thrust, MMH/MON

4: High Thrust, MMH/MON

5: Medium Thrust, MMH/MON

6: Medium Thrust, Hydrazine/MON

Second Trade-Off: Results

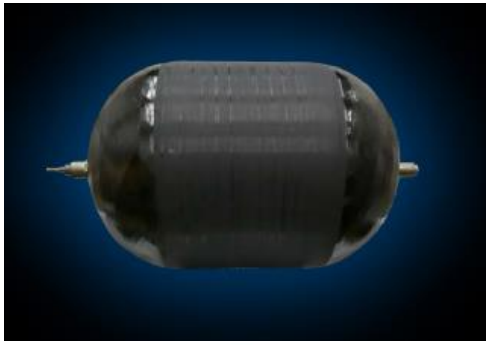




AEROJET ROCKETDYNE

	Main Engine	Thruster
Propellant	High Thrust, MMH/NTO	
Model	R-42	R-6D
Thrust [N]	890	22
Isp [s]	303	294
O/F	1,65	1,61
Mass [kg]	4,53	0,454
Producer	Aerojet Rocketdyne	

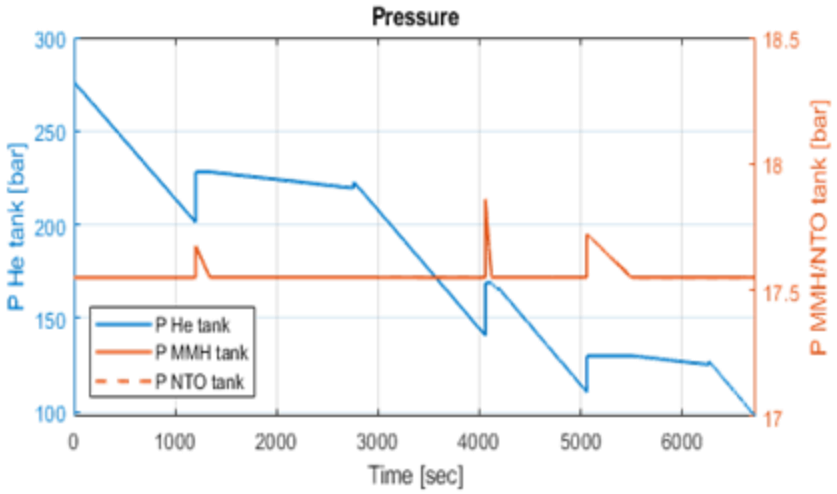
Propellant and Pressurant Tank Selection

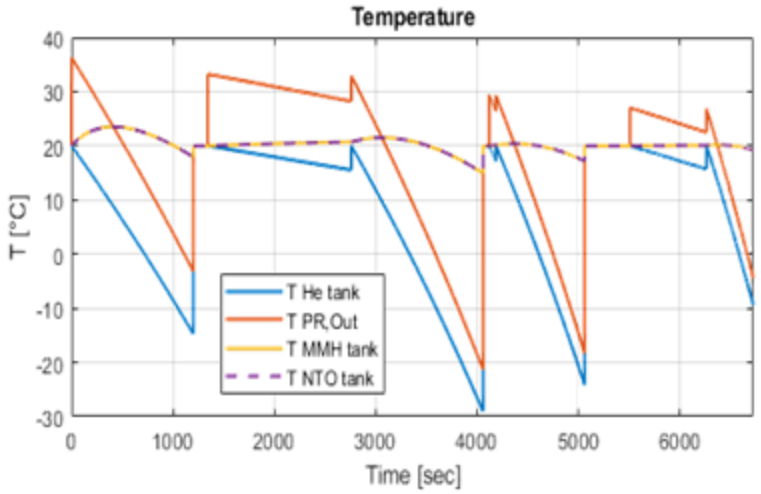


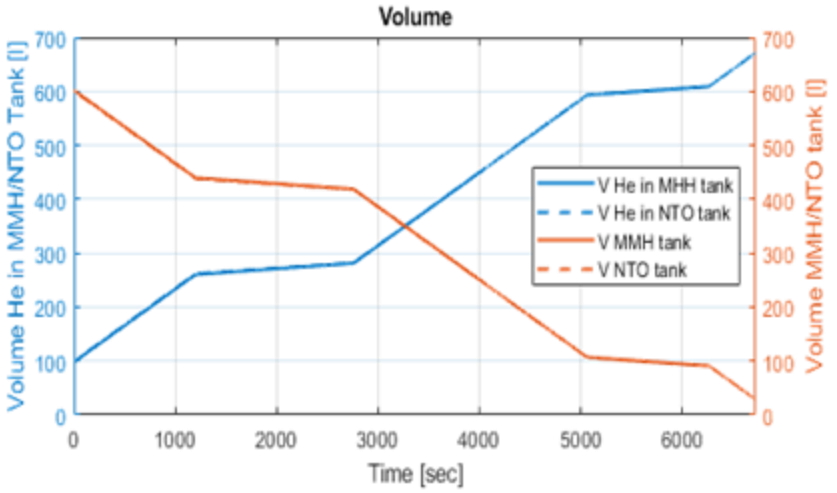


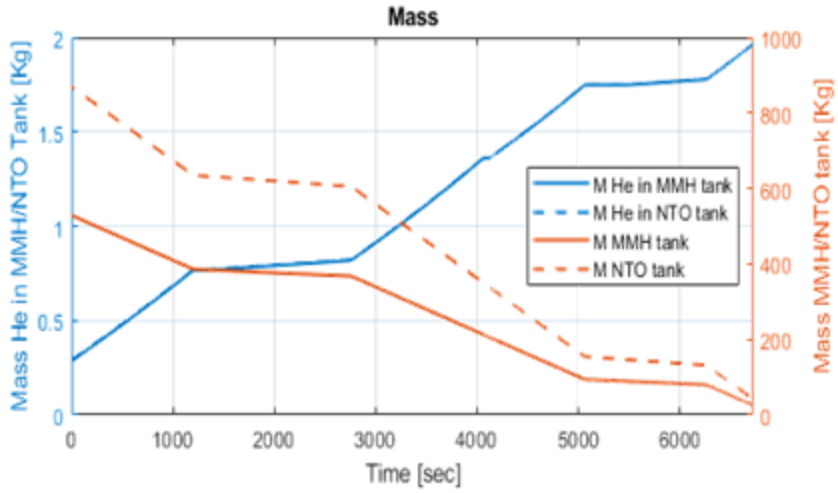
	MMH & NTO Tank	He Tank
Model	OST 22/X	PVG 120
Volume [L]	700	120
Mass [kg]	36	23,5
MEOP [bar]	19,5	310
Producer	MT Aerospace	Ariane Group

Tank Modelling

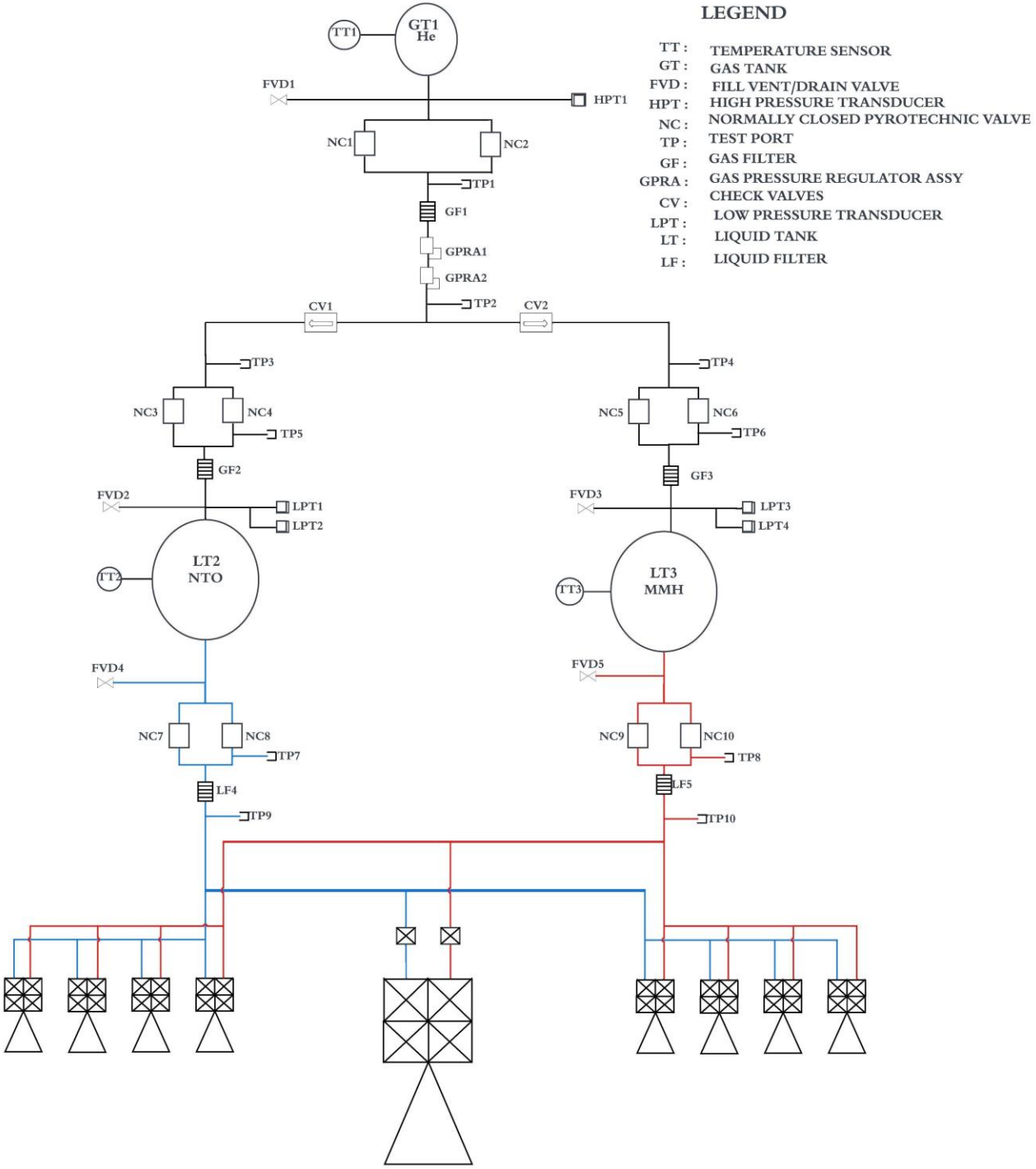








P&ID


LEGEND
TT : TEMPERATURE SENSOR
GT : GAS TANK
FVD : FILL VENT/DRAIN VALVE
HPT : HIGH PRESSURE TRANSDUCER
NC : NORMALLY CLOSED PYROTECHNIC VALVE
TP : TEST PORT
GF : GAS FILTER
GPRA : GAS PRESSURE REGULATOR ASSY
CV : CHECK VALVES
LPT : LOW PRESSURE TRANSDUCER
LT : LIQUID TANK
LF : LIQUID FILTER
TT1
GT1 He
FVD1
HPT1
NC1
NC2
TP1
GF1
GPRA1
GPRA2
TP2
CV1
CV2
TP3
TP4
NC3
NC4
NC5
NC6
TP5
TP6
GF2
GF3
FVD2
FVD3
LPT1
LPT2
LPT3
LPT4
TT2
LT2 NTO
TT3
LT3 MMH
FVD4
FVD5
NC7
NC8
NC9
NC10
TP7
TP8
LF4
LF5
TP9
TP10

List of Equipment

Main Components		
Item	**Count**	**Mass [kg]**
Propellant Tank	2	72
Helium Tank	1	23,5
Thruster	8	3,632
Main Engine	1	4,53

Total Mass [kg]	
w/o margins	117,521
with margins	**124,190**

Additional Components		
Item	**Count**	**Mass [kg]**
Temperature Transducer	3	0,6
Fill & Drain Valve/Test Port	5/10	0,45/0,9
H. P./ L.P Transducer	1/4	0,125/0,5
NC Pyrovalve	10	1,6
Liquid/Gas Filter	2/3	0,36/0,072
Check Valve	2	0,272
P. Regulator	2	3,6
Latching Valves	2	0,68
Pipe Lines	-	5

Mass Budget

	Start Mass [kg]	EOL Mass [kg]
MMH	529	25
NTO	870	41
Helium	6	6
Dry Mass	665	665
Wet Mass	2070	737
Launch Mass	2155	

Outlook

- Complete Propulsion System Design
- 445 kg of Buffer
- 100% COTS Components
- Further Work: Testing and Integration

Thank you for your attention!



Any Questions?

Annex: First Trade-Off: Evaluation

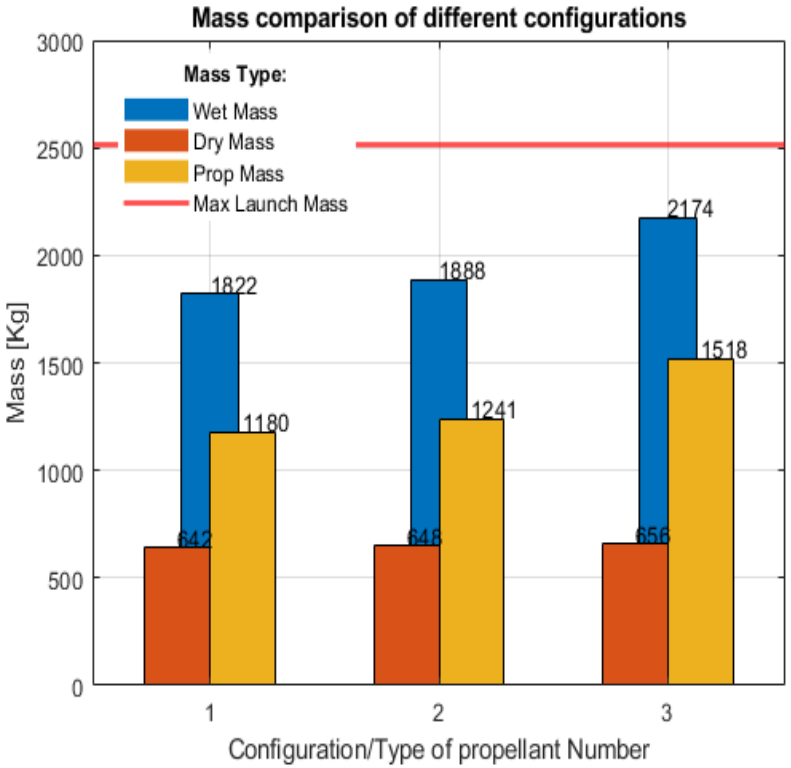


Fig.1. Wet mass, propellant mass, and dry mass of each configuration.

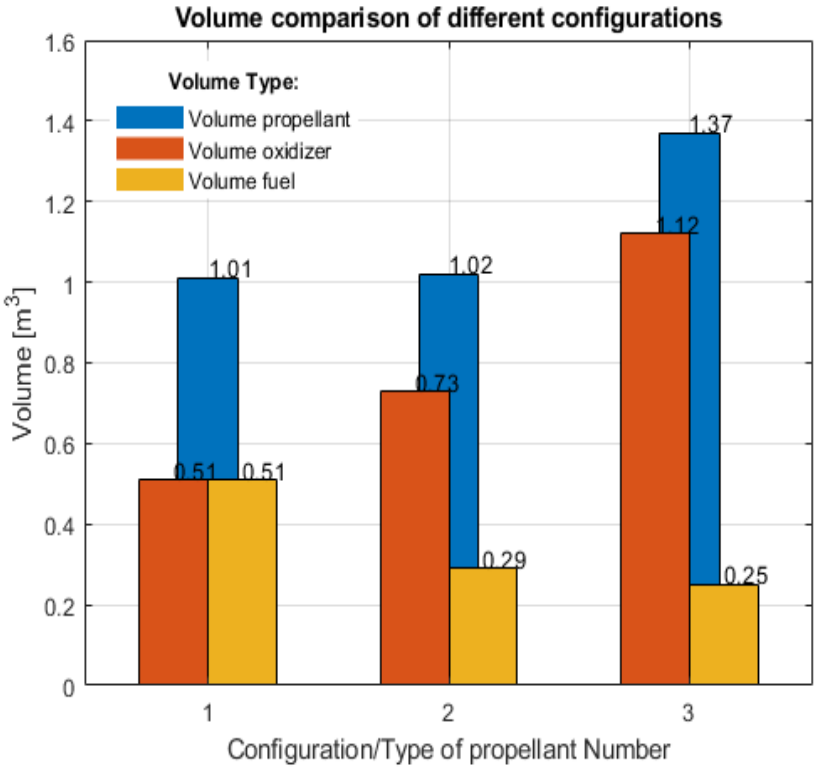


Fig.2. Propellant volume, fuel volume, and oxidizer volume of each configuration.

Annex: **Second Trade-Off:** Configurations

Main Engine	MMH/NTO		MMH/MON			Hydrazine/MON
Config.	1	2	3	4	5	6
Thrust [N]	425	890	425	1000	420	450
Isp_{vac} [s]	321	305	320,6	321	319,5	329
Producer	Ariane	Aerojet	Aerojet	Nammo	Nammo	IHI

Thrusters	MMH/NTO		MMH/MON			Hydrazine/MON
Config.	1	2	3	4	5	6
Thrust [N]	22	22	22	22	22	22
Isp_{vac} [s]	294	294	302	302	302	295
Producer	Aerojet	Aerojet	Aerojet	Moog	Moog	IHI

Annex: Second Trade-Off: Evaluation

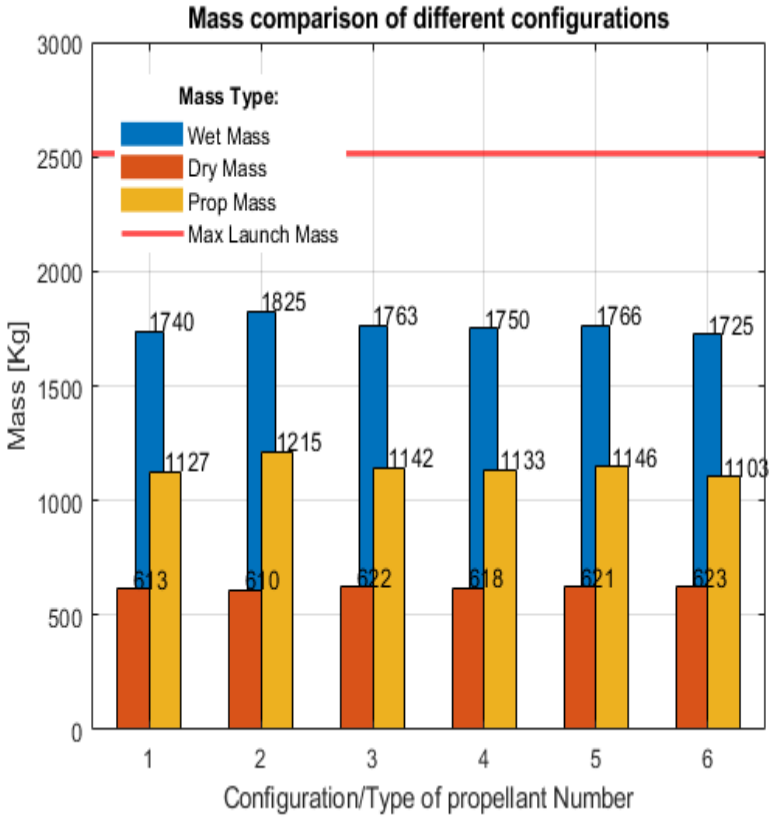


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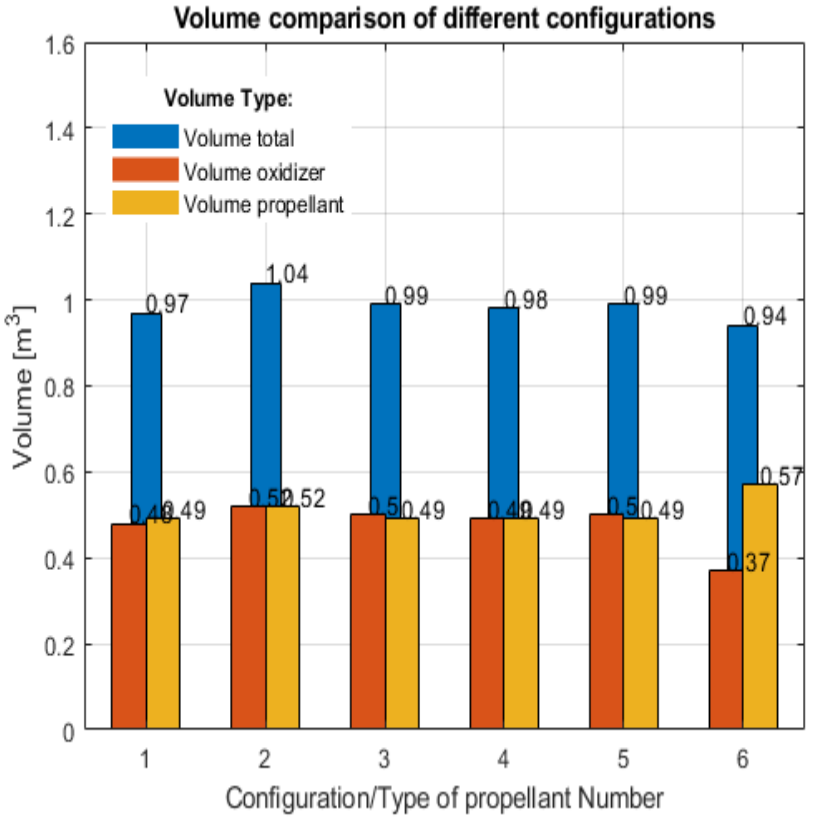


Fig.2. Propellant volume, fuel volume, and oxidizer volume of each configuration.